Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years building **React** front ends and Python and Node.js **APIs** on **SQL** databases, writing **automated tests** with every change, with booking and payments experience and eagerness to grow in .NET Core.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized relational **data models** on Azure SQL and PostgreSQL with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through indexing and query tuning.
- Built FastAPI services and integrations with data validation, error handling, and reconciliation checks, shipping through automated **CI/CD** releases and running 5K+ monthly jobs at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack platform with **React** views for uploads, grading history, and question management on Flask REST APIs, serving 150+ instructors and 10K+ daily requests with 60% faster feedback.
- Implemented Flask-Login **authorization**, secure PDF, DOCX, and ZIP upload handling, and GitHub Actions CI/CD to Render and Vercel, processing 1K+ files per batch with fault-tolerant error handling.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ **unit and integration test** suites with PyTest and Mockito and automated nightly Jenkins runs across Windows, Linux, and macOS, generating coverage reports and failure logs in Agile sprints.
- Debugged and optimized existing large-scale **API integrations** with pagination, field filtering, and caching, improving response times 30% and reducing technical debt 20% through object-oriented Singleton and Factory refactoring.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a MERN platform with **React** and eight Node.js microservices for authentication, billing, payments, and notifications, syncing 2M+ records across locations via REST APIs and real-time WebSockets.
- Automated Jest, Mocha, and Selenium unit, integration, and end-to-end tests in Jenkins **CI/CD** with Docker staging on Azure App Services, reaching 95%+ build validation and monitoring with Sentry.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Python Flask APIs for bookings, time-slot **reservations**, pricing, and coupons with Firebase real-time sync and transactional locking to prevent double-booking, sustaining sub-200ms latency across 50K+ booking records.
- Integrated multi-currency PayPal, card, UPI, and wallet **payments** with idempotent webhooks for refunds, disputes, and chargebacks, cutting manual payment handling by 70% and speeding booking confirmations 40%.

Skills

Languages: JavaScript, TypeScript, Python, SQL, Java, HTML, CSS
Frontend: React, Next.js, Angular, TailwindCSS, Component Testing
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, Data Modeling
Testing & Quality: Jest, Mocha, Selenium, PyTest, Mockito, Unit Testing, Integration Testing, Code Reviews
Cloud & Delivery: Azure, AWS, Docker, Git, GitHub Actions, Jenkins, CI/CD, Sentry, Stripe, Agile/Scrum

Projects

HMIS – Hospital Management Information System Live | *TypeScript, MERN, Twilio, Sentry, TailwindCSS*

- Won Smart India Hackathon with a TypeScript and **React** booking app for appointment scheduling, rescheduling, and cancellation with Twilio confirmations and Sentry monitoring across 100+ daily appointments.

PiXMod Live | *Next.js, Stripe, MongoDB, CLIP, Stable Diffusion*

- Built **Stripe** subscription billing and premium access flows in a Next.js SaaS app with MongoDB storage and AI image tools, lifting premium conversion 18% while processing 1K+ edited assets.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*

- Built a TypeScript and Next.js platform with real-time WebSocket updates, advanced transaction filtering, and a token-based firewall, deployed via **CI/CD** on Kubernetes for 50K+ users and 1M+ transactions.